Ajay Kumar Dwivedi

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Profile

SQL Server Database Performance Engineer and Database Reliability Engineer with 15+ years of hands-on experience operating large SQL Server databases in 24x7 production environments on on-premises servers, SQL Server on Azure VMs and AWS EC2, Azure SQL Managed Instance, and Azure SQL Database. I identify performance bottlenecks, tune queries and execution plans using Query Store, Extended Events, DMVs, and wait statistics, and build automation in T-SQL, PowerShell, Python, and Terraform. I also support PostgreSQL on Patroni, RDS, and Aurora, and I work closely with application teams through root cause analysis and SLA-driven on-call support.

Core Competencies

- **SQL Server Performance Engineering:** Performance Tuning, Performance Engineering, Query Tuning, Query Optimization, Query Store, Extended Events, Execution Plan Analysis, Wait Stats, Index Tuning, Performance Bottlenecks, Deadlock Analysis, Parameter Sensitivity, Memory Grants, Stored Procedure Tuning, Plan Regression, SQL Server Profiler, Database Internals
- **SQL Server Administration:** SQL Server Administration, Always On Availability Groups, Failover Clustering, Log Shipping, Transactional Replication, Change Data Capture (CDC), Backup and Restore, Integrity Checks, Index Maintenance, Table Partitioning, Database Health Checks, Patching and Upgrades, Ola Hallengren Maintenance Solution, Database Design Patterns
- **Azure SQL and Cloud:** SQL Server on Azure Virtual Machines, Hosted SQL Server, IaaS, PaaS, Azure SQL Database, Azure SQL Managed Instance, Amazon RDS for SQL Server, SQL Server on AWS EC2, Multi-Cloud, DBaaS, Cloud Migration
- **High Availability, DR and Incidents:** High Availability and Disaster Recovery, Business Continuity, Database Encryption, Transparent Data Encryption (TDE), Always Encrypted, Incident Management, Priority Incident Management, Root Cause Analysis, ITIL, 24x7 Operations, 24x7x365 Operations, On-Call Rotations, SLA Commitments, Operational Standards
- **T-SQL Development and Standards:** T-SQL, Stored Procedures, Dynamic SQL, Code Review, Design Reviews, Technical Standards, Database Best Practices, Source-Controlled Deployments, SQL Database Projects, DACPAC
- **Security and Access Control:** Database Security, Least Privilege, Access Control, Auditing, Vulnerability Assessment, Credential Management, Identity and Access Management
- **Automation and Scripting:** Infrastructure as Code, Terraform, Ansible, PowerShell, Python, T-SQL, Automation Frameworks, CI/CD, GitHub Actions, Deployment Automation, Capacity Planning, Cost Optimization
- **Monitoring and Observability:** Observability, Monitoring, Alerting, Grafana, Prometheus, Datadog, PagerDuty, Dashboards, SQL Server Profiler, Database Mail
- **Collaboration and Leadership:** Application Team Support, Stakeholder Management, Customer Service, Written and Verbal Communication, Attention to Detail, Self-Starter, Taking Initiative, Problem Solving, Technical Leadership, Mentoring

Skills

- **High Availability and Disaster Recovery:** Clustering, Availability Groups, Mirroring, Log Shipping, Replication, Disk Stripping, RAID, Multi-Subnet Setup
- **Databases:** SQL Server, Azure SQL Database, Azure SQL Managed Instance, PostgreSQL, RDS, Aurora, SQLite
- **Cloud Platforms:** Azure, AWS, GCP

- **Programming Languages:** T-SQL, SQL, PowerShell, Python, JavaScript, HTML, CSS, Shell Scripting
- **Tools and Frameworks:** Git, Linux, Ansible, Flask, GitHub Actions, VS Code, Grafana, Prometheus, Containers, Docker, Ubuntu, RHEL, PowerShell DSC, Infrastructure as Code (IaC)
- **Languages:** English (proficient), Hindi (native)

Experience

Tessell *February 2026 - September 2026*

Principal Solution Architect - SQL Server

- Built DBaaS capabilities for deploying SQL Server on AWS EC2 and Azure VMs, enabling self-service concurrent deployments
- Wrote Python-based SAGA workflows on Kubernetes and a dockerized environment to build Tessell's Database as a Service (DBaaS) platform, a distributed systems architecture with Docker and Kubernetes as the control plane and AWS, Azure, and GCP as the data plane, reducing provisioning time to 20 minutes
- Added monitoring for data and log file growth using SQLExporter, Prometheus, and VictoriaMetrics
- Added enhancements to support Change Data Capture (CDC) and transactional replication
- Used root cause analysis to debug production issues and implement fixes for customers during 24x7 on-call support
- Designed CentOS-to-RHEL migration automation for PostgreSQL using Python and Ansible, migrating 180+ database clusters with 15 minutes of downtime each
- Provided technical leadership on database architecture for Tessell's DBaaS SaaS platform, covering provisioning, high availability, and day 2 operations, and mentoring engineers

AngelOne *February 2022 - February 2026*

Senior Site Reliability Engineer (DevOps)

- Tuned SQL Server queries by analyzing execution plans to fix forced serialization, scalar UDF inlining, lazy spools, cardinality misestimates, wrong join types, and residual I/O performance issues, performance bottlenecks, and deadlocks
- Diagnosed SQL Server plan issues using Query Store, Extended Events, DMVs, and wait statistics, including memory grants, spills, parameter sensitivity, and plan regressions
- Established repeatable performance baselines, reviewed workload history to tune access paths, bind variables, concurrency, and justified query hints, and rewrote stored procedures to improve plan stability
- Performed SQL Server administration across production instances, including patching, upgrades, and configuration, and ran database health checks and statistics maintenance, reporting findings and remediating issues
- Reviewed T-SQL code with code review guidance and managed source-controlled deployments for SQL Server releases
- Established database best practices and technical standards for SQL Server configuration, backups, and security, and delivered internal training on them
- Implemented single-click automation for Transparent Data Encryption (TDE) certificate rotation, helped customers implement Always Encrypted, upgraded SSL and TLS drivers, and built a credential manager for identity and access management, applying least privilege to service accounts while securely storing and retrieving their credentials for DBA automation
- Implemented database security controls for SQL Server instances, including encryption, access hardening, and auditing of privileged access to meet regulatory compliance requirements
- Automated common database team activities, such as SQL Server installation, configuration, monitoring, and alerting through Database Mail, using PowerShell, Flask, Python, T-SQL, Grafana, and PagerDuty for on-premises and cloud Managed Instances
- Automated AWS database services across IaaS and PaaS, including RDS for SQL Server, deployment

and maintenance using Terraform
- Built the SQLMonitor tool with dashboards that supported capacity planning and estimated storage needs for the next couple of years based on past growth
- Created a GitHub repo-based approach to baselining all Windows Server and Linux servers, simplifying database monitoring with Prometheus and Grafana
- Provided production support for SQL Server and PostgreSQL databases on a 24x7 on-call rotation, meeting SLA commitments for availability and incident response
- Owned business continuity for database clusters, maintaining and testing failover and recovery procedures
- Handled request management for database access, change, and provisioning requests within the team's incident and change workflow
- Developed runbooks, knowledge base articles, and documentation for database operations, enabling on-call engineers to resolve incidents
- Automated multi-datacenter HA and standalone PostgreSQL setup using Ansible, saving 80 hours per week of SRE effort with a single-click GitHub Actions workflow for cluster deployment and configuration
- Led rolling upgrades of PostgreSQL clusters with minimal downtime, using Patroni-managed failover

Arcesium *May 2020 - February 2022*
Database Reliability Engineer

- Improved database reliability by automating ScriptOut of entire SQL Server instances for disaster recovery
- Applied reliability engineering as a Database Reliability Engineer, upgrading 30 critical SQL clusters and AGs in 12 hours to avoid a week-long downtime
- Built single-click audit infrastructure for databases so application teams can set up auditing and generate reports on table usage, data modifications, security events, and data changes
- Resolved hundreds of incidents by moving 38 TB of AG databases with PowerShell
- Built automation to prevent common issues such as tempdbSaver and LogSaver
- Administered PostgreSQL databases on on-premises servers and AWS RDS and Aurora, covering high availability, backups, and upgrades

TiVo *February 2018 - May 2020*
Database Administrator

- Built a resilient backup strategy with Ola Hallengren's maintenance solution, including native backup and restore, integrity checks, and recovery testing, automated scheduled refreshes of non-prod environments, set up log shipping from one prod server to another, and enabled single-click full recovery of a SQL Server instance or database
- Reduced transactional replication issues by 80% using latency monitoring infrastructure
- Reduced performance incidents by 80% by building Delta Index Defragmentation automation for replicated and HADR databases
- Implemented a change management process using SQL Database Projects, SQL DACPAC, and GitHub Actions
- Saved 40+ hours by automating SQL Server deployment and configuration using PowerShell
- Implemented monitoring and alerting with Datadog

United Health Group *October 2016 - February 2018*
Senior SQL DBA

- Improved performance issues by 50% and reduced VLDB maintenance time
- Handled incident management for production issues, debugging customer issues and implementing fixes during 24x7 on-call support
- Practiced stakeholder management by directly interacting with customers to fix critical customer-facing performance and deployment issues

- Saved $1 million per year through DB space capacity automation

Ericsson Global *June 2015 - October 2016*
Technical Lead

- Achieved a 3x performance improvement through advanced query optimization, such as recursive CTEs and table partitioning
- Mentored 18+ developers in following best practices and guidelines

TATA Consultancy Services *March 2011 - June 2015*
SQL DBA Developer

- Saved 20+ hours per week by automating core DBA tasks using dynamic SQL
- Moved 250+ large tables from large databases, each with billions of records, using SSIS bulk loads in 45 hours
- Built data warehouse ETL pipelines using Pentaho and Kettle for high-volume liquor retail stores

Projects

SQLMonitor *January 2021 - Present*
Open-source, enterprise-grade SQL Server monitoring tool (GitHub)

- Built using T-SQL, PowerShell, Python, and Grafana
- Monitoring 500+ SQL Servers across environments
- Sends observability alerts to Slack, PagerDuty, and email
- Built-in auto-actions to clear and resolve issues

SQLDBATools *July 2021 - February 2023*
PowerShell module with cmdlets for SQL Server DBA activities (GitHub)

- Covers server inventory, disk and space estimation, backup history, permission audits, and maintenance job setup, reducing manual DBA effort by 60% across 120+ servers
- Built a monitoring and alerting system with Grafana dashboards, auto-clearing alerts, email notifications, and alert suppression
- Published to PSGallery for installation with `Install-Module`

AI Agent *October 2025 - Present*
LLM-powered SQL agent that answers natural-language performance questions about SQL Server instances monitored by SQLMonitor (GitHub)

- Built using Python, LangChain, Ollama, Streamlit, and SQLMonitor
- Grounds every answer in SQL queries against SQLMonitor data, shows the query in each explanation, and includes prompt guardrails against write statements, enabling application teams to troubleshoot performance problems themselves

HadrSync *January 2021 - Present*
PowerShell module that synchronizes all objects across replicas in Availability Groups (GitHub)

- Ensures zero sync issues during failovers and DR events
- Detects and synchronizes delta changes, such as changes to passwords, definitions, and more

Certifications

- Microsoft Certified Professional - Database Querying (October 2013)
- Microsoft Certified Professional - Database Administration (December 2013)
- BrentOzarUnlimited Mastering SQL Server (November 2022)
- SQLSkills Blackbelt Professional (June 2023)

Education

Bachelor of Engineering in Electronics and Communication Engineering *July 2006 - June 2010*
RustamJi Institute of Technology, Gwalior, Madhya Pradesh

Awards and Honors

- **Tech Titan**, AngelOne for 10x performance optimization on production SQL Server (2022)
- **Sapphire**, United Health Group for innovation and automation (2017)

Availability

Notice Period: 7 Days, Open to Relocation, Open to remote opportunities